Non-linear Modeling and PID Control of Twin Rotor MIMO System

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***Abstract*— This paper deals with PID control tuning for a non-linear multi-input multi-output system (MIMO). In this paper, a laboratory helicopter model called the twin rotor MIMO system (TRMS) is considered as a MIMO system. The mathematical modeling of helicopter model is simulated using MATLAB/Simulink for controlling it. The two degrees of freedom (2-DOF) are controlled both in horizontal and vertical direction using proportional-integral-derivative (PID) controller. Simulation results are obtained for different input signals. Moreover the error of the plant output and control output of PID controller are presented which show the performance evaluation of TRMS.**

Keywords- Multi-input multi-output (MIMO), twin rotor MIMO system (TRMS), proportional-integral-derivative (PID).

I. INTRODUCTION

TRMS is designed in MATLAB/ Simulink which has high-order, non-linear, MIMO system with significant cross coupling [1]. Its behavior resembles that of a helicopter in certain aspects. The TRMS is the laboratory setup designed for control experiments. The prime control objective is to balance the TRMS in coupled condition and to make the beam of the TRMS to follow the desired trajectory or to reach desired positions in 2-DOF accurately and quickly. This is done by using PID control technique. In this paper, the performance of TRMS is observed by using both simple PID controller and cross coupled PID controller. The simple PID controller controls the vertical and horizontal movements separately. In simple PID control system, influence of one rotor on the motion in the other plane is not compensated by the controller structure whereas in cross coupled control system it is controlled [2]. However the system is not decoupled. The proposed cross coupled control structure consists of four PID controllers in which the inputs are independent to each other.

The gain parameters of the PID controller decide the system performance. Simulations results of TRMS system performance are obtained and tabulated for both simple and cross coupled PID controller to analyze the difference in system output error and control output of PID controller.

II. SYSTEM DESCRIPTION

The TRMS mechanical structure consists of two rotors placed on a beam together with a counterbalance whose arm with a weight at its end is fixed to the beam at ht e pivot and it determines a stable equilibrium position. The entire structure is attached to the tower allowing for safe helicopter control experiments. In normal helicopter, angle of attack is a control for controlling the aerodynamic force whereas in the laboratory setup as shown in Fig.1, the angle of attack is fixed. Hence, the speeds of rotors are varied to control the aerodynamic force. Therefore, supply voltages of dc motors are changes to control the rotation speed of the propeller which will change the corresponding position of the beam, where ω and ψ are the corresponding horizontal and vertical positions of the beam [3].

Usually, the phenomenological models are nonlinear, that means at least one of the positions of the beam is an argument of a nonlinear function. In order to present such a model as a transfer function, it has to be linearised. According to the electrical-mechanical diagram of TRMS shown in Fig. 1, the non-linear model equations can be derived.

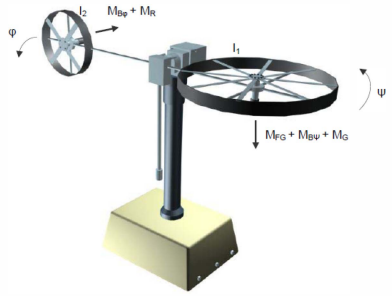


Fig 1. TRMS phenomenological model

As far as the mechanical unit is concerned the following momentum equations can be derived for the vertical movement [4], [5], [6]:

$$I_1 *(d^2\psi/dt^2) = M_1 - M_{FG} - M_{B\psi} - M_G \quad (1)$$

where

$$M_1 = a_1 * \tau_1^2 + b_1 * \tau_1 \text{ –non linear static characteristic} \quad (2)$$

$$M_{FG} = Mg * sin\ \psi \text{ –gravity momentum} \quad (3)$$

$$M_{B\psi} = B_{1\psi} * (d\psi/dt) + B_{2\psi} * sign\ (d\psi/dt)$$

-friction forces momentum (4)

$$M_G = K_{gy} * M_1 * (d\psi/dt) * cos\ \psi \text{ –gyroscopic momentum} \quad (5)$$

The motor and the electrical control circuit is approximated by a first order transfer function thus in Laplace domain the motor momentum is described by,

$$\tau_1 = (k_1/(T_{11}s + T_{10})) * u_1 \quad (6)$$

Similar equations refer to horizontal plane motion,

$$I_2 *(d^2\varphi/dt^2) = M_2 - M_{B\varphi} - M_R \quad (7)$$

where

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$M_2 = a_2 * \tau_2^2 + b_2 * \tau_2$ –non linear static characteristic (8)

$M_{B\psi} = B_{1\varphi} * (d\psi/dt) + B_{2\varphi} * sign\ (d\varphi/dt)$

-friction forces momentum (9)

$M_R = (k_c\ (T_0 s+1)/\ (T_p s+1)) * \tau_1$ (10)

Again the DC motor with the electrical circuit is given by

$\tau_2 = (k_2 / (T_{21}s+T_{20})) * u_2$ (11)

III. Implementation of TRMS Model Using Cross Coupled PID Conroller

The simple PID controller controls the vertical and horizontal movements separately. In this control system, influence of one rotor on the motion in the other plane is not compensated by the controller structure. The system is not decoupled. Equations (1) to (11) and with the TRMS model parameters [7], the mathematical modeling of TRMS is designed using simple PID controller. The horizontal and vertical movement plant output are compared with desired position output using comparator and then the error is processed to simple PID controller as shown in Fig. 2. Then the controlled output of simple PID controller is given to both of the plants to obtain the desired position response.

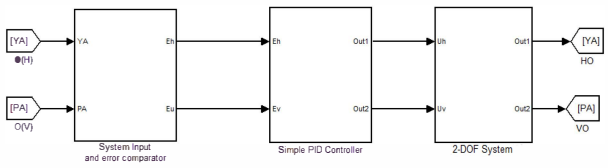


Fig 2. Mathematical modeling of TRMS using simple PID controller

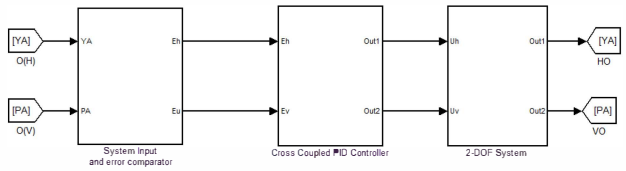


Fig 3. Mathematical modeling of TRMS using cross-coupled PID controller

The TRMS mathematical model using cross coupled PID controller is implemented in MATLAB/Simulink exactly as shown in Fig. 3. The PID_H, PID_{HV}, PID_{VH} and PID_V are the four PID controllers used here to control the system in the vertical and horizontal planes as shown in Fig. 4. The subscripts indicate the source-sink relation of the controller. In this control system, influence of one rotor on the motion in the other plane can be compensated by the cross coupled structure of controller [8], [9]. Each control signal (U_v and U_h) is the sum of two controller outputs. The saturation block is used to limit the absolute value of controls. E_h and E_v are the horizontal and vertical error which is the difference between the reference signal and system output.

IV. Simulation Results and Discussion

For tracking the desired trajectories of TRMS, the three different reference inputs are given to the system. They are step wave, sine wave and square wave. The Fig. 5 shows the system outputs of TRMS in 2 DOF using simple PID control. Table I shows the results of the simple PID control of the TRMS. The error of the step response is 109.49 in horizontal plane whereas the error in vertical plane is 60.69. So, the proposed controller has less error in vertical plane compared to the horizontal plane. Table II shows the results of cross-coupled PID control of TRMS. In this control, the error of the step response is 150.91 in horizontal plane and the error in vertical plane is 63.55. Moreover the error of square response and control output in vertical plane is 67.05 and 890.56 whereas in [10] the error of square response is 112.85 and 656.37. So, in the proposed cross coupled PID control of TRMS as shown in Fig. 4, the better control of error in vertical plane is obtained.

In simple PID control, the TRMS has good tracking performance both in horizontal and vertical plane when the reference input is sine wave as shown in Fig. 5 (c) and (d). The control output in horizontal and vertical plane is 197.81 and 555.95. The error of sine wave is 31.66 and 18.97 in horizontal and vertical plane whereas the error in cross coupled control of TRMS in vertical plane is 14.53. Hence, it has better control than simple PID control in vertical plane when the reference input is sine wave. The Fig. 6 shows the system outputs of 2 DOF using cross-coupled PID control.

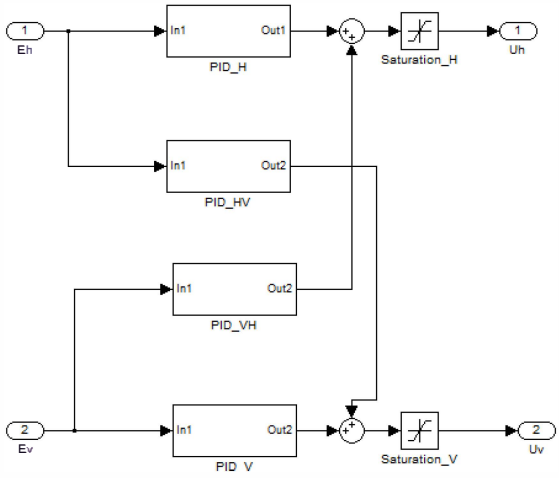


Fig 4. Internal control structure of cross-coupled PID controller

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TABLE I
SIMULATION RESULTS FOR 2-DOF CONTROL USING SIMPLE PID CONTROLLER

Plane	Reference	Error	Control output
Horizontal	Step	109.49	163.30
	Sine	31.66	197.81
	Square	170.37	264.24
Vertical	Step	60.69	891.81
	Sine	18.97	555.946
	Square	80.83	844.394

TABLE II
SIMULATION RESULTS FOR 2-DOF CONTROL USING CROSS COUPLED PID CONTROLLER

Plane	Reference	Error	Control output
Horizontal	Step	150.91	288.00
	Sine	39.31	231.15
	Square	220.81	364.49
Vertical	Step	63.55	891.35
	Sine	14.53	532.74
	Square	67.05	890.56

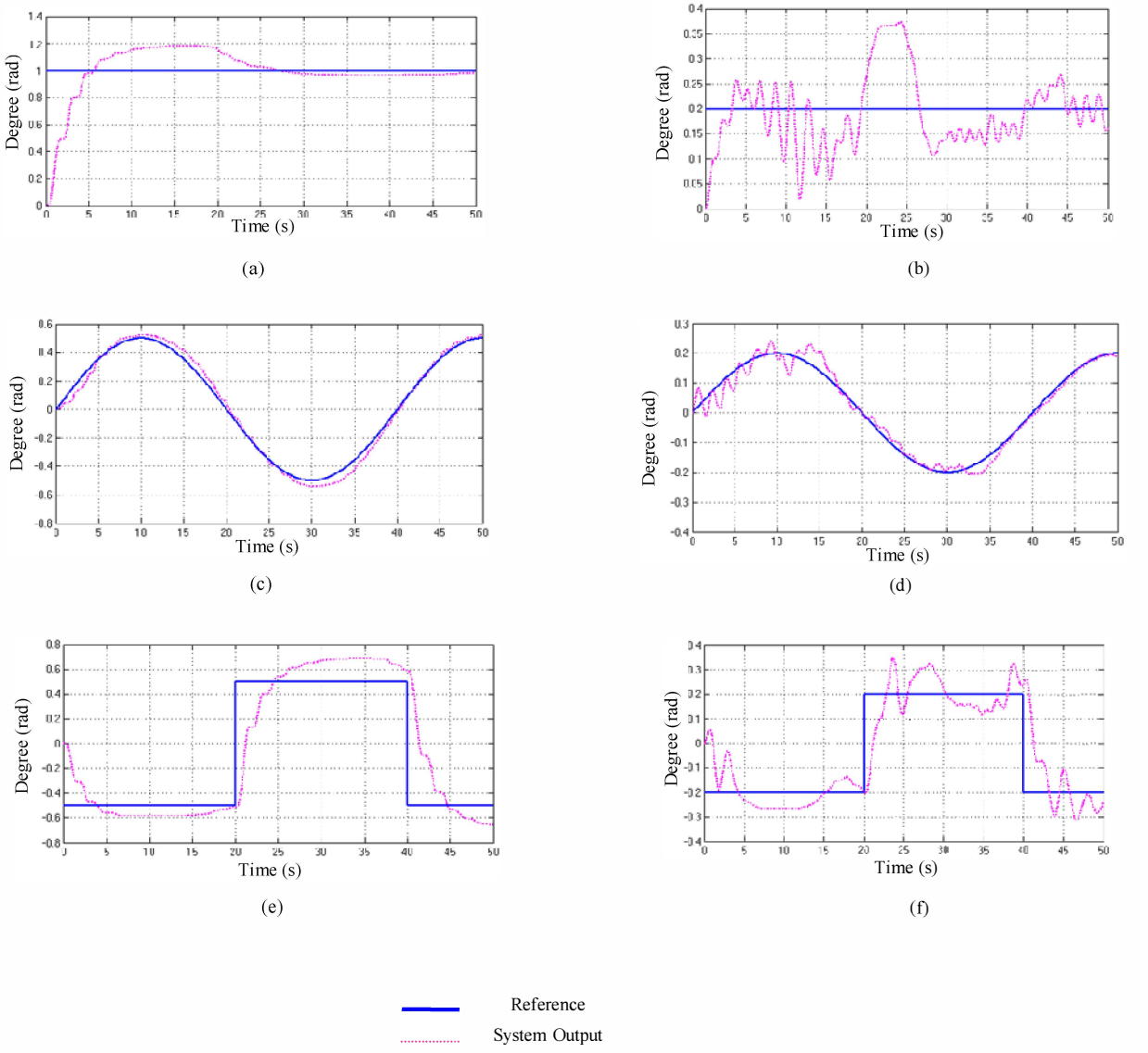


Fig 5. System outputs of 2 DOF using simple PID control. (a) and (b) are Step response in horizontal and vertical plane. (c) and (d) are Sine wave response horizontal and vertical plane. (e) and (f) are Square wave response horizontal and vertical plane.

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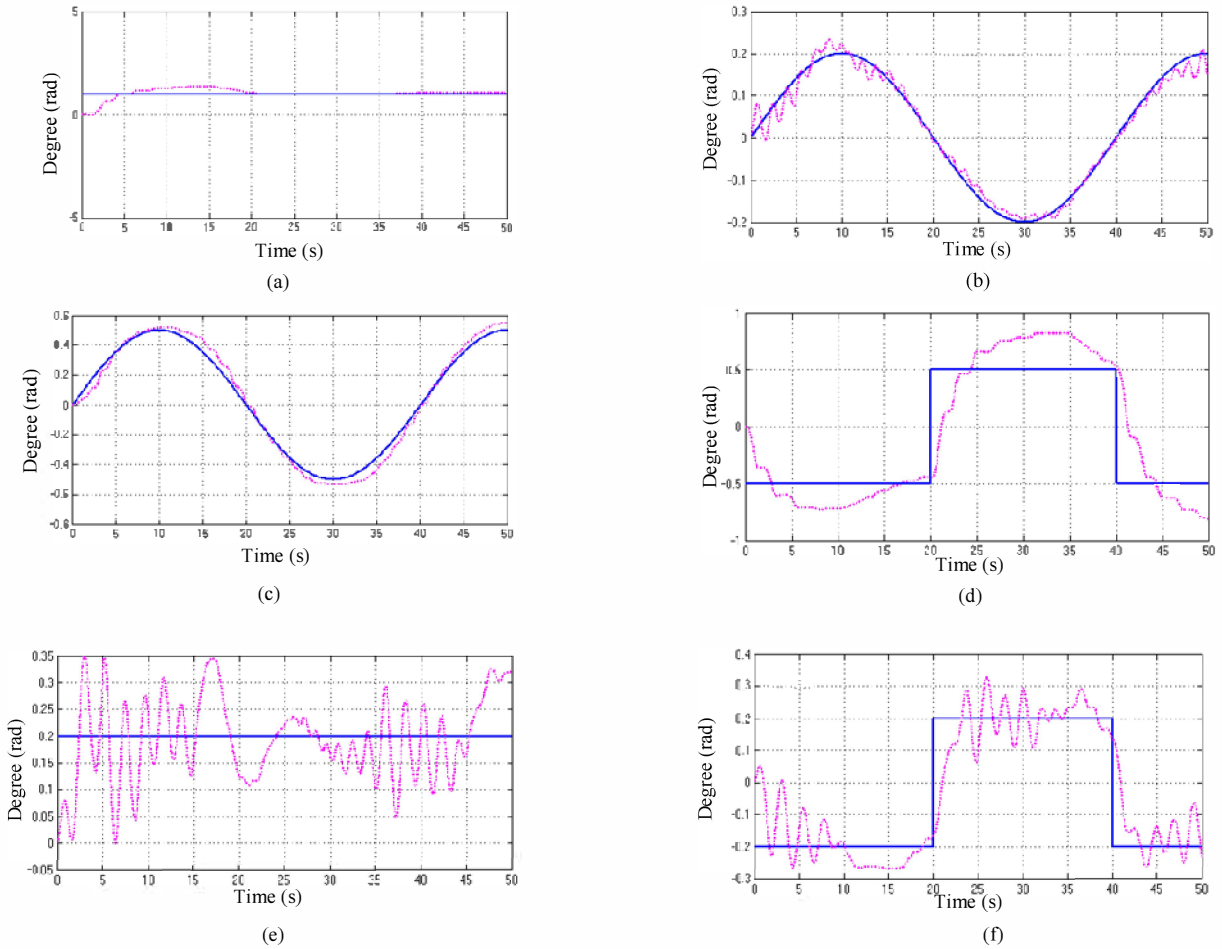


Fig 6. System outputs of 2 DOF using cross coupled PID control. (a) and (b) are Step response in horizontal and vertical plane. (c) and (d) are Sine wave response horizontal and vertical plane. (e) and (f) are Square wave response horizontal and vertical plane.

V. Conclusion

In this paper, to analyze and control the system non-linearity, the mathematical modeling of TRMS in 2-DOF is designed using MATLAB/Simulink. The simple and cross coupled PID controllers are implemented for the control and simulation results are obtained. The process output error and PID control outputs are evaluated for different reference inputs and the total error of sine wave is reduced.

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